HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

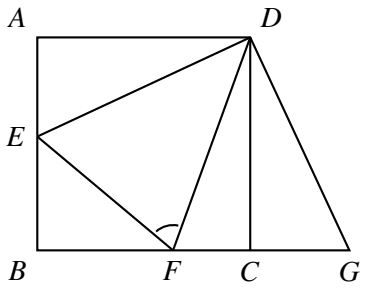
Student No.: 16 Name: _____ Marks: / 18

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

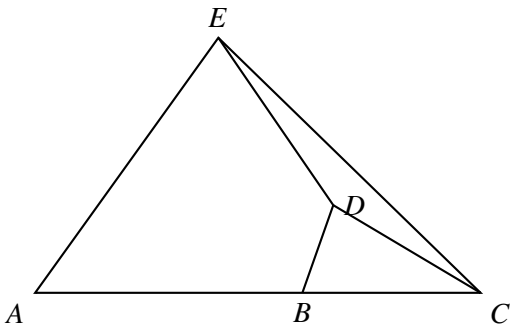
(DSE-2014 PII Q16)

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2. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

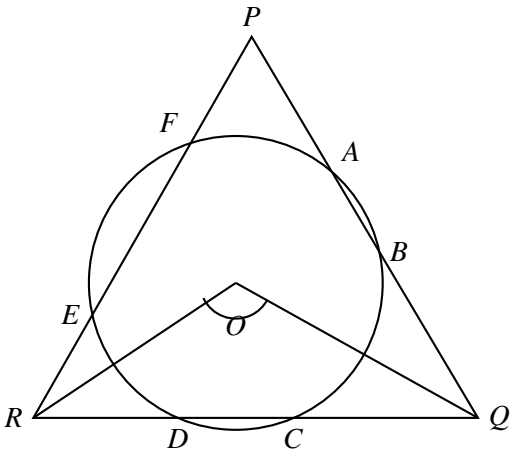
(DSE-2014 PII Q17)

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3. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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4. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

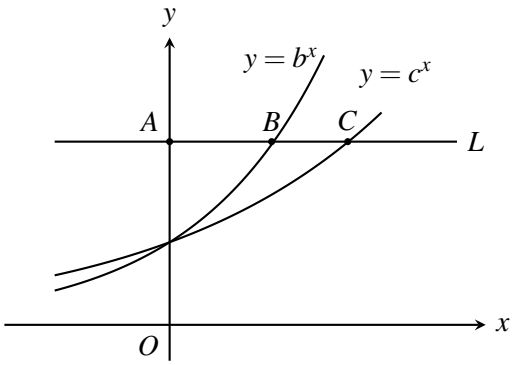
(DSE-2014 PII Q23)

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5. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

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6. If $m > 1$, which of the following are geometric sequences?

I. $2^m, 2^{2m}, 2^{3m}, 2^{4m}$

II. $m, 2m^2, 3m^4, 4m^8$

III. $\log m, \log m^2, \log m^4, \log m^8$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q37)

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7. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

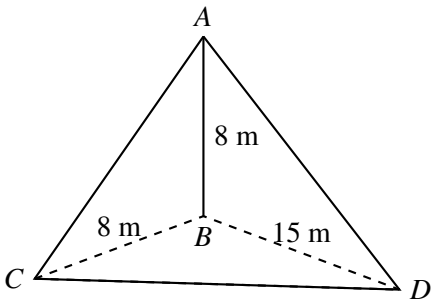
(DSE-2014 PII Q39)

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8. In the figure, AB is a vertical pole standing on the horizontal ground BCD, where $\angle CBD = 90°$. If the angle between the plane ACD and the horizontal ground is θ, then $\tan\theta =$



A. $\frac{8}{15}$.

B. $\frac{15}{8}$.

C. $\frac{15}{17}$.

D. $\frac{17}{15}$.

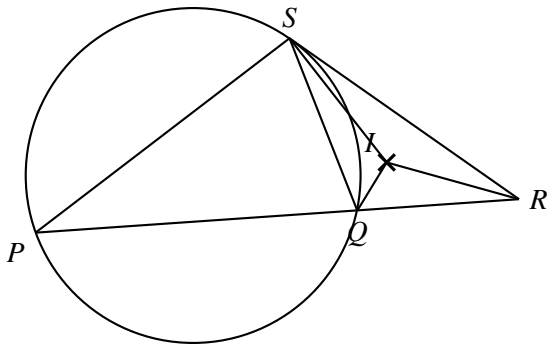
(DSE-2014 PII Q40)

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9. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

I can finish _____ questions

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